EENG 385 - Electronic Devices and Circuits
PID Contoller: Plant Characterization
Lab Handout

Outcome and Objectives

The PID Controller board is designed as a practical laboratory experience for students to understand how to characterize a plant, and tune a PID Controller. Through this process you will achieve the following learning objectives:

- Provide students will learn the “feel” of how PID control works by changing the P, I, D terms of the controller and noting the output.
- Building a Bode plot for a plant (low pass filter).
- Build a Nyquist plot for a plant (low pass filter).
- Using the Nyquist plot, predict the gain at which the system goes unstable.

The PID Controller Board

A high-level overview of the PID Controller board is shown in Figure 1. The input can come from a push-button (step input), a potentiometer (analog input), or a function generator using a user selectable jumper (high tech, I know). The Kp, Ki and Kd terms are adjustable from 0 to infinity using a potentiometer with a long stem. The control signal is converted into a PWM signal whose duty cycle is proportional to the control signal. This PWM signal is then converted into an analog output using a low pass filter.

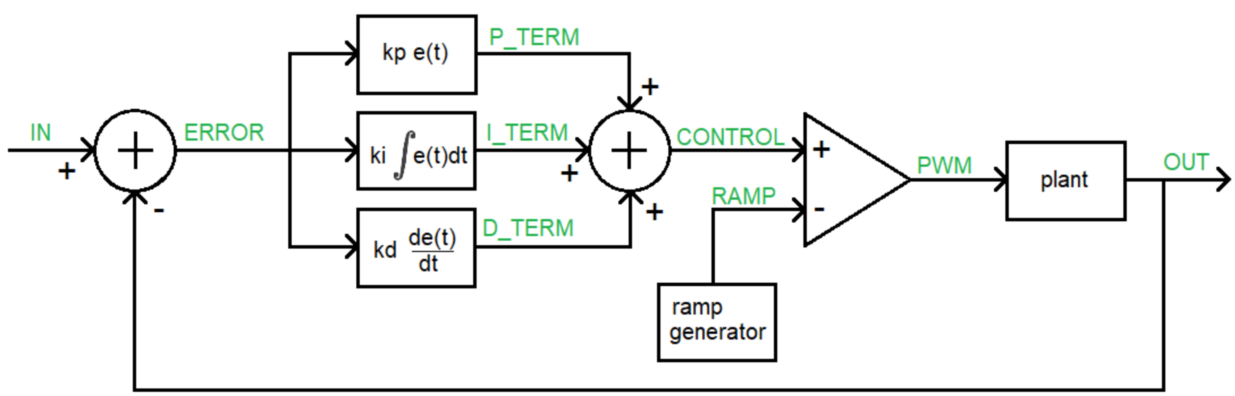
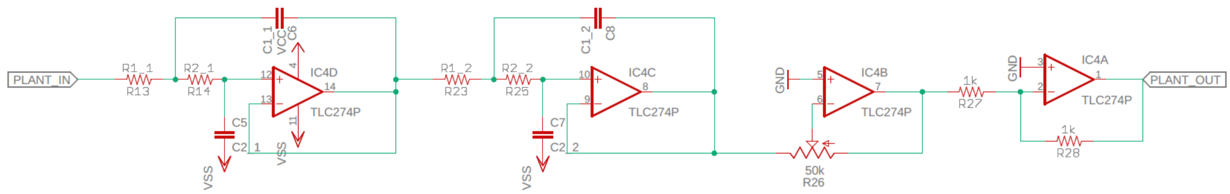


Figure 1: High-level architecture of the PID Controller board.

Importantly, the input and output of the plant can be isolated from the controller circuit. This will allow you to characterize the plant using the FRA capability of our oscilloscopes. Well now that you have an idea of what you are working with, let’s talk more about the plant.

The Plant:

The plant of the PID Controller board is a 4th order low pass filter built using opamps IC4D/C The gain of the plant is adjusted using the trimmer POT R26 attached to opam IC4B. Since this opamp is inverting, the final opamp is an inverting configuration with gain -1.



In order to accommodate different plants the LPF components do not have fixed values, instead they use labels which are populated by components whose values create different filter types shown in the table below. Note, the filters were designed using the Analog Devices, Analog Filter Wizard using the following parameters.

	-3dB Chebyshev	-1dB Chebyshev	Butterworth	Bessel
Pass	-3dB @ 100Hz	-3dB @ 100Hz	-3dB @ 100Hz	-3dB @ 100Hz
Stop	-40dB @ 300Hz	-40dB @ 300Hz	-40dB @ 400Hz	-40dB @ 500Hz
Resp	-3dB Chebyshev	-1dB Chebyshev	Butterworth	Bessel
Opt	Voltage Range	Voltage Range	Voltage Range	Voltage Range
Capacitor	E6	E6	E6	E6
Resistors	E6	E6	E6	E6
R1_1	68kΩ	33kΩ	15kΩ	10kΩ
R2_1	100kΩ	68kΩ	22kΩ	15kΩ
C1_1	100nF	100nF	100nF	100nF
C2_1	22nF	33nF	68nF	68nF
R1_2	150kΩ	100kΩ	22kΩ	10kΩ
R2_2	330kΩ	150kΩ	100kΩ	22kΩ
C1_2	100nF	100nF	100nF	100nF
C2_2	680pF	1.5nF	10nF	33nF

Your board will have one of these four different amplifier configurations. Take a moment now, examine the components in your board and determine which LPF is populating your board.

The User Interface:

Let's look at the ways that you can configure the PID Controller board shown in Figure 2 to make measurements and control the plant.

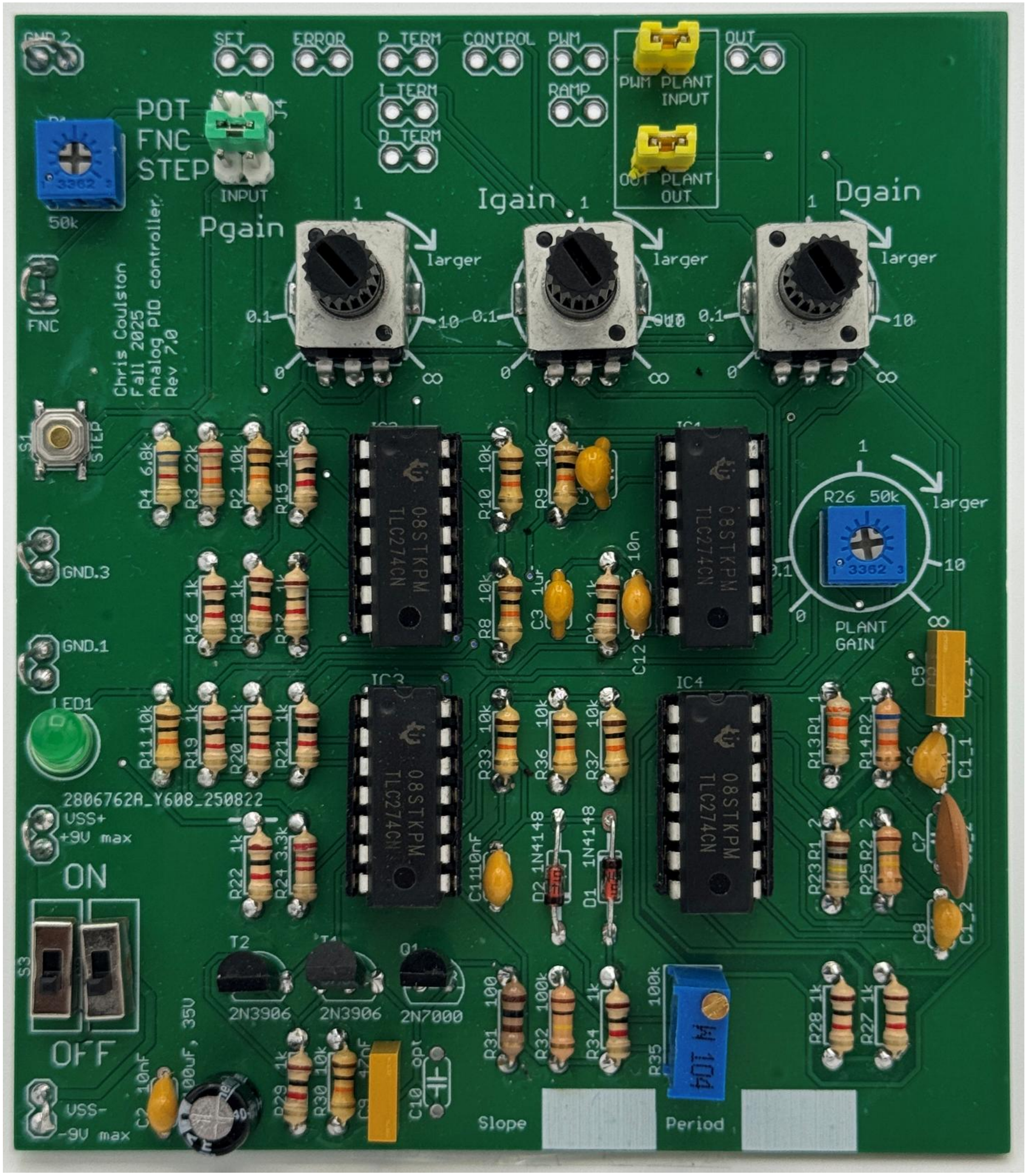


Figure 2: An assembled PID Controller board.

Power Input

Use the 3 wires loops in the lower left corner of the PCB to connect power.

- The bottom loop, labeled VSS- is connected to -9V
- The middle loop labeled VSS+ is connected to +9V
- The upper loop labeled GND.1 is connected to GND

Each ON/OFF switches turns on +9V or -9V separately. I did this because we have SPDT switches in stock and do not have DPDT switches. Always throw both switches at the same time, either both ON or both off. This is simple to do with a single finger.

Input Selection

In the upper left corner of the board is a 2x3 male header with a green jumper installed. The green jumper can connect one of three inputs to the PID controller input

1) An analog input provided by the blue potentiometer just to the left of the header.
2) A step input actuated by the push button on the left edge of the board labeled “STEP”
3) A function generator that connects to the loops labeled FNC and GND.2

As shown in Figure 2 the jumper is connecting the function generator loop to the input of the PID controller.

Plant Selection

At the top right of the board is a pair of yellow jumpers

Plant Gain

The plant gain is controlled by the blue potentiometer on the right side of the PCB and labeled PLANT GAIN. This is a single turn trimmer POT so will be very sensitive to small changes in the position. In FIGURE 1, the potentiometer is positioned just a bit shy of 1. You should adjust the POT so that the cross used to adjust the position of the center tp is as close to 1 as you can get.

PID Gains

The gain for each term is controlled by the Pgain, Igain or Dgain POTs in the top center of the board. When set to their mid position the gain of each term is 1. Full CCW corresponds to a gain of 0 and full CW, a gain of infinity.

Oscilloscope probe points

These are the set of unpopulated header pins at the top of the PCB. They are shown in FIGURE 4 above a block diagram of a PID controller. Each of the terms is a voltage that, with the exception of the PWM and RAMP signals, that represents the signed magnitude of the signal. So for example if the error is -1V, then the corresponding error signal is -1. When the plant is included in the feedback loop and the PID values tuned properly, you should see the OUT voltage track the SET voltage closely.

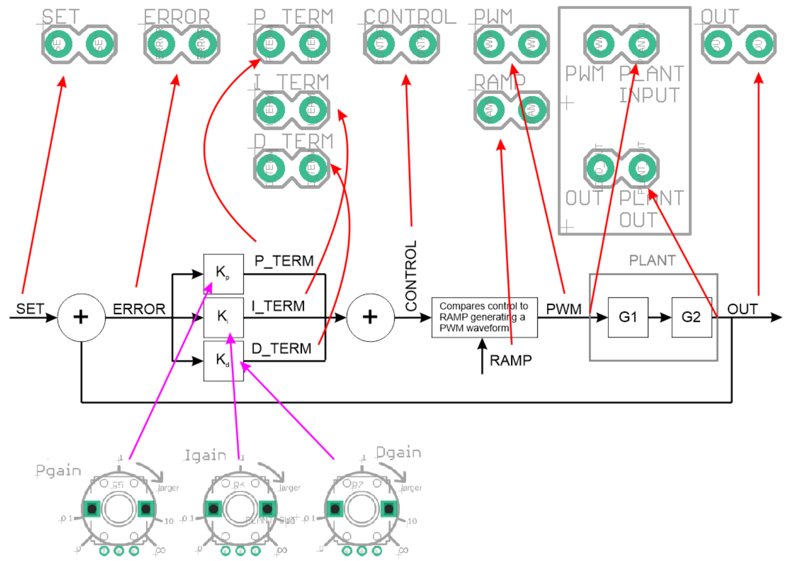


Figure 3: The headers at the top of the board provide access to the values of the PID control loop using an oscilloscope.

Characterize The Plant:

The first thing that you will want to do is characterize the plant by making a Bode Plot. The goal of this will be to first adjust the plant gain so that it is 0dB in the pass-band and the second, to collect numerical data at this gain to build the Nyquist plot of the plant in Excel.

To do this you will:

1) Isolate the plant by removing the input and output jumpers.
2) Power on the oscilloscope, they take forever to boot.
3) Turn on and configure the power supply to have 9V, 0.1A on channel 1 and 2. Leave the power supply on, but **disable** the individual channels until everything is connected.
4) Configure the power supply to generate +9V, GND, -9V, attach the respective terminals to the PID Controller using red (+9V), green (GND) and black (-9V) banana cables and alligator clips.
5) Run a T-connected BNC from the oscilloscope function generator output to Channel 1 of the oscilloscope and to the PLANT_INPUT inputs on the PID Controller board.
6) Attach a probe to Channel 2 of the oscilloscope and connect it to the PLANT_OUT. No need for a ground clip, the function generator ground has already taken care of that.
7) Enable channel 1 and 2 of the power supply and ensure that you are not hitting the current limit on either channel. If so, **immediately disable both channels**. You have something incorrectly connected. Fix and try again.

Here is what your setup should look like. Yea, it's messy ☹

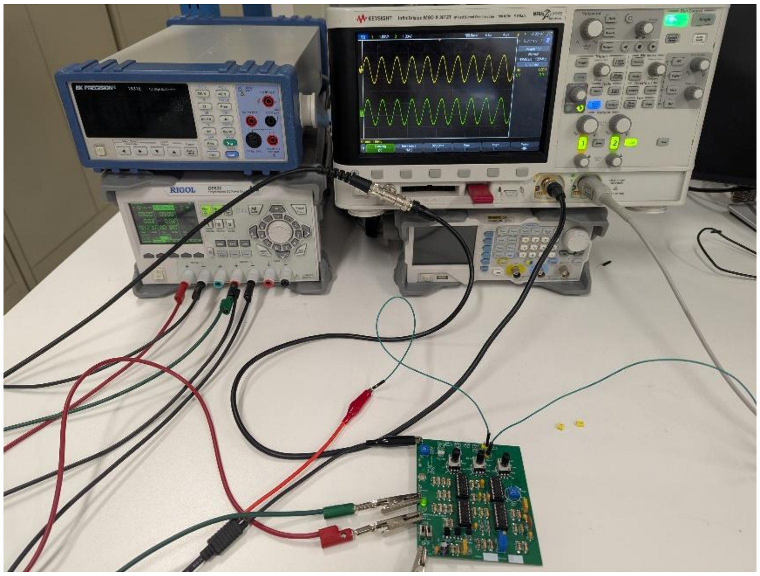


Figure 4: The PID Controller board configured to characterize the plant. Channel 2 is connected to the PLANT OUT off screen to the right.

Before running FRA on the oscilloscope, you should enable the function generator output for a frequency well inside the corner frequency, say 10Hz. Make sure that the input and output have the same amplitude (one probe is not 10x off), you should see a slight phase lab on the output like that in Figure 4. Now adjust the plant gain until the output has the same amplitude as the input, 0dB. Cool.

Before going on, examine the step response of the plant to a 4Hz 2V square wave. Without knowing that your plant is 4th order, would you characterize the plant's step response behavior as first order or second order? Why? Grab a screen shot of the step response to include in your lab report.

Now run the frequency response analysis on the oscilloscope from 10Hz to 1kHz. Your FRA must be close to 0dB through the pass band, 10Hz to 100Hz. If it is -20dB or +20dB then you have an attenuation issue with your scope probes. Figure 5 shows the plot that I got using a -1dB Chebyshev LPF.

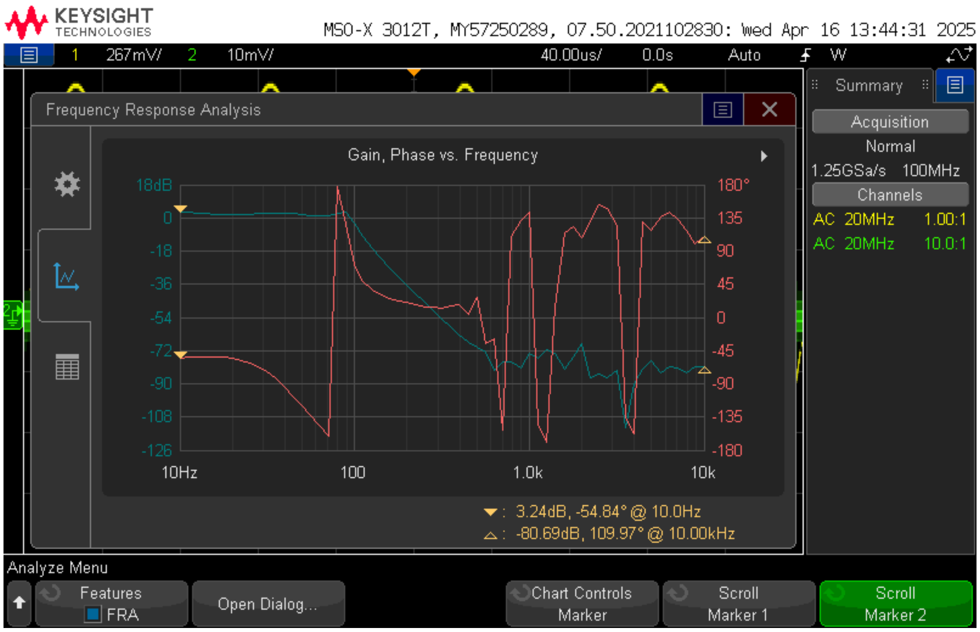


Figure 5: The Bode plot from a -3dB Chebyshev LPF.

Now save the data data used to generate the Bode plot on the oscilloscope by plugging in a thumb drive, pressing the SAVE/RECALL hard button, select the Format option, <u>scroll to the bottom of the options</u>, select **Frequency Reponse Analysis data (*.csv)** and then press the Press to Save softkey. You will use this data in the next section.

Creating a Bode plot

Note that Figure 6 shows additional columns H-K used to build the Nyquist plot in the next section. For now, ignore these columns.

1) Open the .csv file in Excel or import the .csv data into Excel.
2) Immediately save the file as a .xlsx file
3) Delete column C, it just tells you the amplitude of the voltage input.
4) Insert a new column E which converts any positive angles into the using the excel equation =IF(D2<0,D2,D2-360). You can type this equation into cell E2, hit enter, select cell E2, then click on the dot in the lower right corner of the cell and, while holding down the mouse button, drag the cell down, selecting all the rows with data in them. The equation should be copied into all these cells.

	A	B	C	D	E	F	G	H	I	J
1	#	Frequency	Gain (dB)	Phase (°)		Gain (V/V)		X	Y	Flip
2	1	10	2.62	-55.34		1.35207		0.76893	-1.11214	1.11214
3	2	10.8	2.3	-54.69		1.30317		0.75323	-1.06343	1.06343
4	3	11.7	2.03	-54.18		1.26328		0.73932	-1.02434	1.02434
5	4	12.6	1.76	-53.79		1.22462		0.72344	-0.98809	0.98809
6	5	13.6	1.54	-53.61		1.19399		0.70837	-0.96116	0.96116

Figure 6: The header of the Excel file used to generate the Bode and Nyquist plots.

5) Now make a Bode plot by selecting column B and C, then click Insert -> Scatter -> Scatter with Smooth lines. Make the horizontal axis logarithmic by carefully double clicking on one of the numerical values on the horizontal axis. In the Format Axis menu that appears on the right side of the screen, select logarithmic. I also like to change the minimum bound to 10 to fill the graph area. Nice. Do repeat the process with columns B and E to make the phase plot. Note the angle can jump around at higher frequencies. You are welcome to fix these few values by adding -360 degrees to them to smooth out the plot. Here is what I produced.

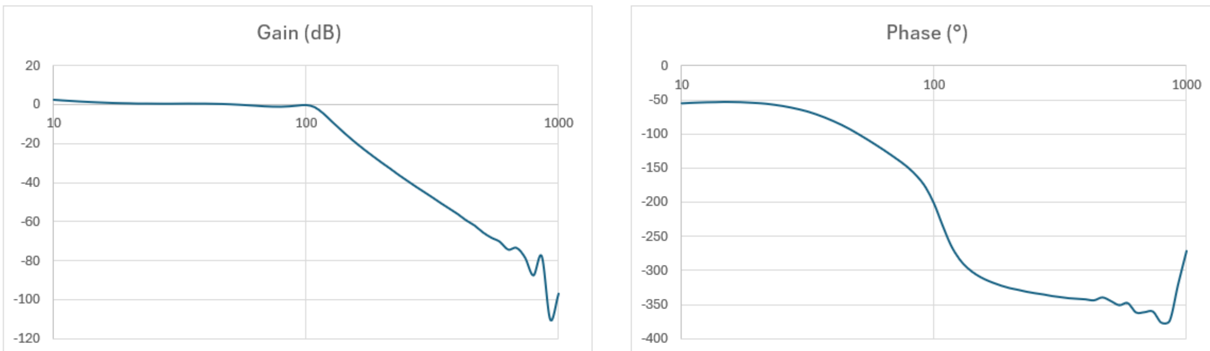


Figure 7: Bode plot for a -1dB Chebyshev LPF with a corner frequency of 100Hz.

Creating a Nyquist plot

1) Insert a header "Gain (V/V) in column G. This column will contain the gain represented as a scaler value. To do this you will have to figure out the equation to convert the dB value in column C into a scaler.
2) Insert a header "X" in column I. This column will contain the x-coordinate of the complex vector that has a magnitude in the G column and a phase angle given in the E column. You will have to use a trig function to accomplish this.
3) Insert a header "Y" in column J. This column will contain the y-coordinate of the complex vector that has a magnitude in the G column and a phase angle given in the E column. You will have to use a trig function to accomplish this.
4) Select columns I and J and then Insert -> X Y (Scatter) -> Scatter with Smooth Lines. You now have half of the Nyquist plot. You can get the other half of the Nyquist plot by mirroring this half around the Y-axis. In other words multiply each Y coordinate by -1.
5) Insert a header "Flip" in column K. This column will contain the negative value in column J. After you populate this column, right click on you half Nyquist plot, click Select Data… In the Select Data Source pop-up, click Add. In the Edit Series pop-up, click in the Series X values box, then drag select all the X data in column I. Click in the Series Y values box, then drag select all the Flip data in column K. Click OK twice and your Nyquist plot should be complete.
6) Take a moment to adjust the X and Y scale of the plot by grabbing the lower right corner of the plot and stretch it so that the X and Y axis' have about the same scale size.

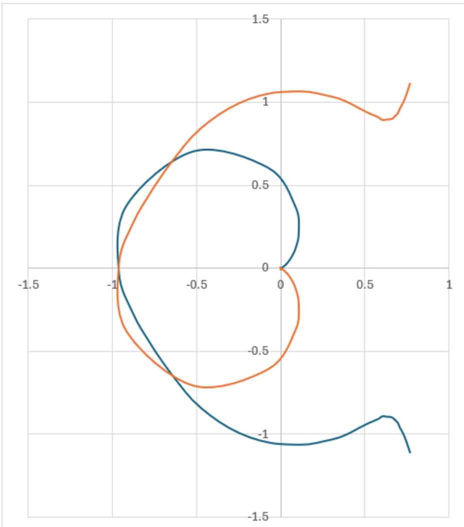


Figure 8: The Nyquist for a 1dB Chebyshev LPF.

Use the Nyquist plot to estimate the gain margin of your plant.

Tune the PID Controller:

Now, lets see if you can tune the PID controller so that the output of the plant can track a 4Hz square wave. To do this you will:

1) Insert the plant into the control loop by installing the jumpers on the PLANT INPUT and PLANT OUTPUT.
2) Power on the oscilloscope, they take forever to boot.
3) Turn on and configure the power supply to have 9V, 0.1A on channel 1 and 2. Leave the power supply on but **disable** the individual channels until everything is connected.
4) Configure the power supply to generate +9V, GND, -9V, attach the respective terminals to the PID Controller using red (+9V), green (GND) and black (-9V) banana cables and alligator clips.
5) Run a T-connected BNC from the oscilloscope function generator output to Channel 1 of the oscilloscope and to the FNC inputs on the PID Controller board. Attach the GND clip of the function generator to the the nearby GND.2 loop.
6) Attach a probe to Channel 2 of the oscilloscope and connect it to the OUT header at the top of the PCB. No need for a ground clip, the function generator ground has already taken care of that.
7) Enable channel 1 and 2 of the power supply and ensure that you are not hitting the current limit on either channel. If so, **immediately disable both channels**. You have something incorrectly connected. Fix and try again.

8) Adjust the oscilloscope function generator to generate a 4Hz square wave with an amplitude of 1V.
9) Enable channel 2, make it the same scale as channel 1 and place the GND references of the two channels in the vertical center of the screen.

Now it's time to play with the PID knobs to get the output to track the input. Figure 9 shows my setup and tuned controller, you should try to do better.

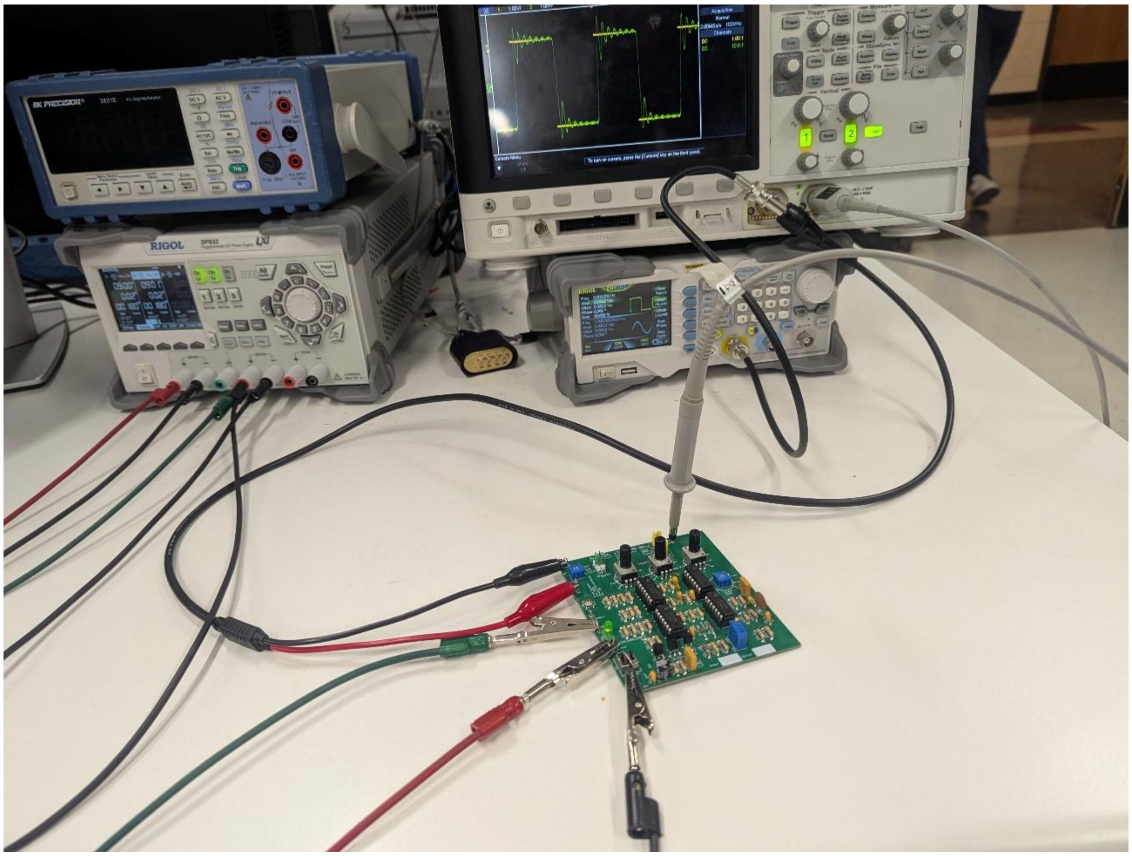


Figure 9: A tuned PID controller that allows the output of a 4th order Chebyshev LPF to track a square wave. Note, I used the function generator, you should probably just use the oscilloscope's function generator.

Screen shot the input/output of your tuned controller. Describe the function of the following controls in two sentences each.

1) The proportional gain
2) The integral gain
3) The derivative gain

Nyquist stability predicts the amount of plant gain that will cause a Kp=1 Ki=0 Kd=0 feedback controller to go unstable. Set you integral and derivative gain to 0 and your proportional gain to 1. My gain margin in Figure 8 was very small, meaning that only the slightest increase in the PLANT GAIN potentiometer should drive my output unstable.

Turn In:

1) Make a record of your response to numbered items below and turn in a single copy as your team's solution on Canvas using the instructions posted there.
2) Include the names of both team members at the top of your solutions.
3) Use complete English sentences to introduce what each of the items listed below is and how it was derived.

LM7805

Questions 1 – 5

Zener Diode

Questions 1 – 3

Power-on LED

Questions 1 – 9

ON/OFF switch

Questions 1 – 2

Reverse Voltage protection

Questions 1 – 7

Assembly

Solder only the components in the power supply area

Operational and tested

- Power-on LED
- 5V
- 2.5V
- Show reverse voltage protection works by applying reverse voltage and verify that the current drawn by the power supply is 0mA